Design and Implementation of an Active Debris Removal System for Mitigating Space Debris in Low Earth Orbit

Group 15

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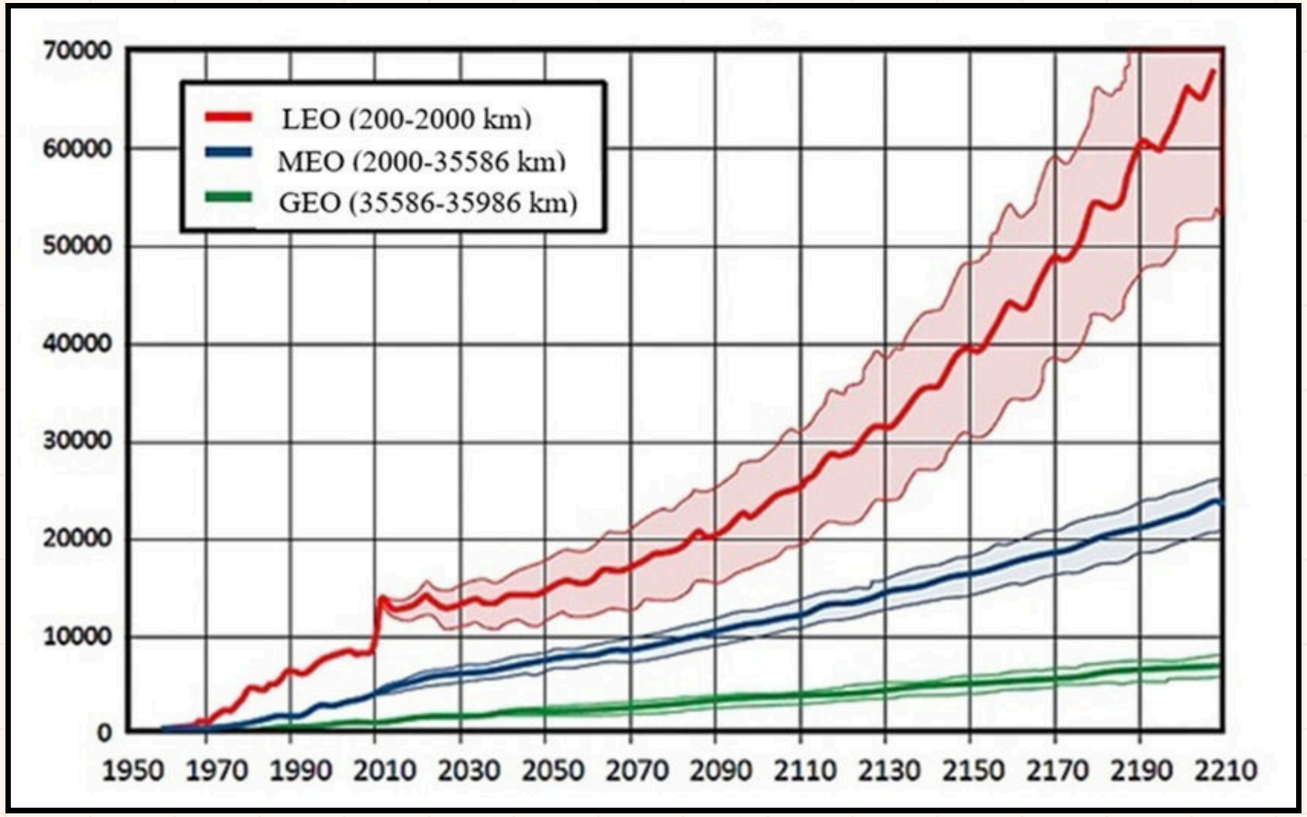
2024/10/10 - Fundamentals of Spaceflight, final presentation

Overview

1. Introduction
2. Removal method
3. Spacecraft sizing
4. Rocket selection
5. Rocket dynamics
6. Earth satellite operations
7. Cost analysis & sustainability
8. Conclusion

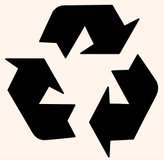
Background

- Satellite crucial for society
- More than 9300 tonnes of debris in orbit
- Might lead to Kessler Syndrome
- Need to remove at least 5 debris/year



Source : ResearchGate

Problem definition

1		**Objective 1** Remove 10 or more debris from LEO, decreasing their decay time to at most 25 years
2		**Objective 2** Mission duration < 1 year
3		**Objective 3** Select an appropriate launch site
4		**Objective 4** Simulate the ascent to target orbit
5		**Objective 5** Perform a sustainability analysis
6		**Objective 6** Perform a cost analysis

Debris selection

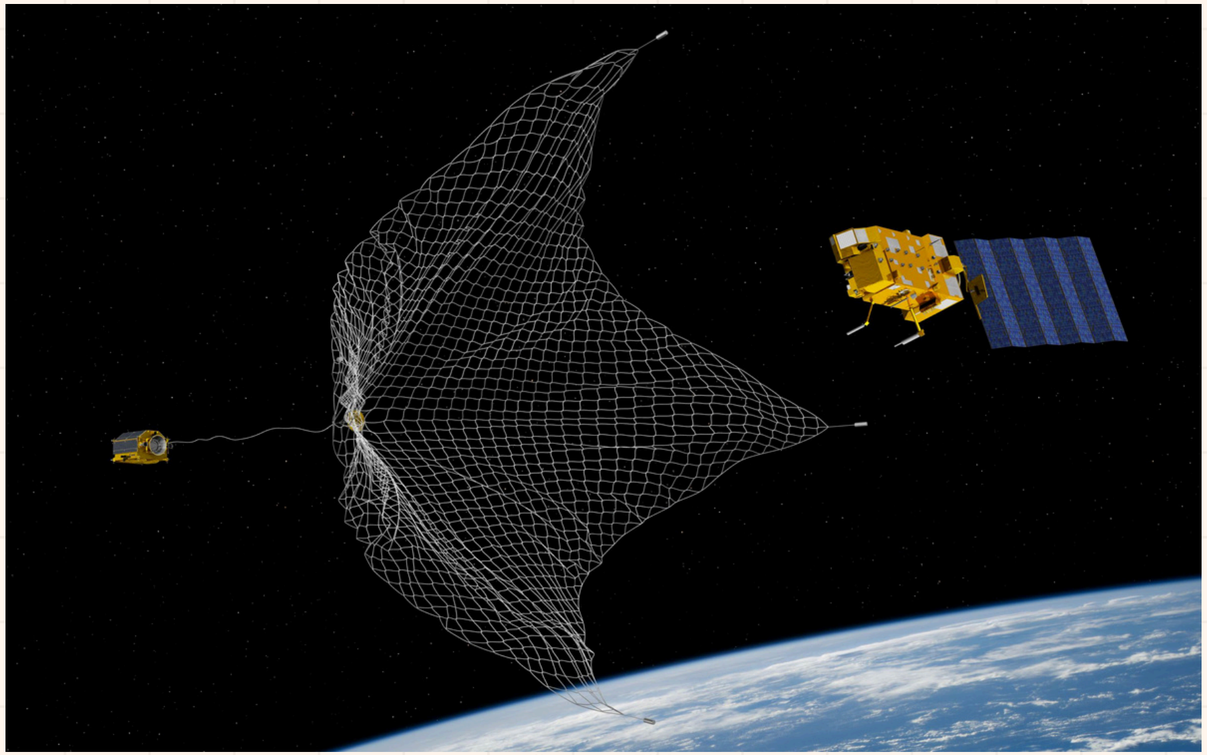
- Used CelesTrak database to find debris
- Search scope :
  - Rocket bodies only
  - Inclination range 95-100°
  - Similar RAAN values
  - Small eccentricity
- 10 adequate debris were found with this scope



Ariane IV second stage,
source : ninfinger.org

Removal method

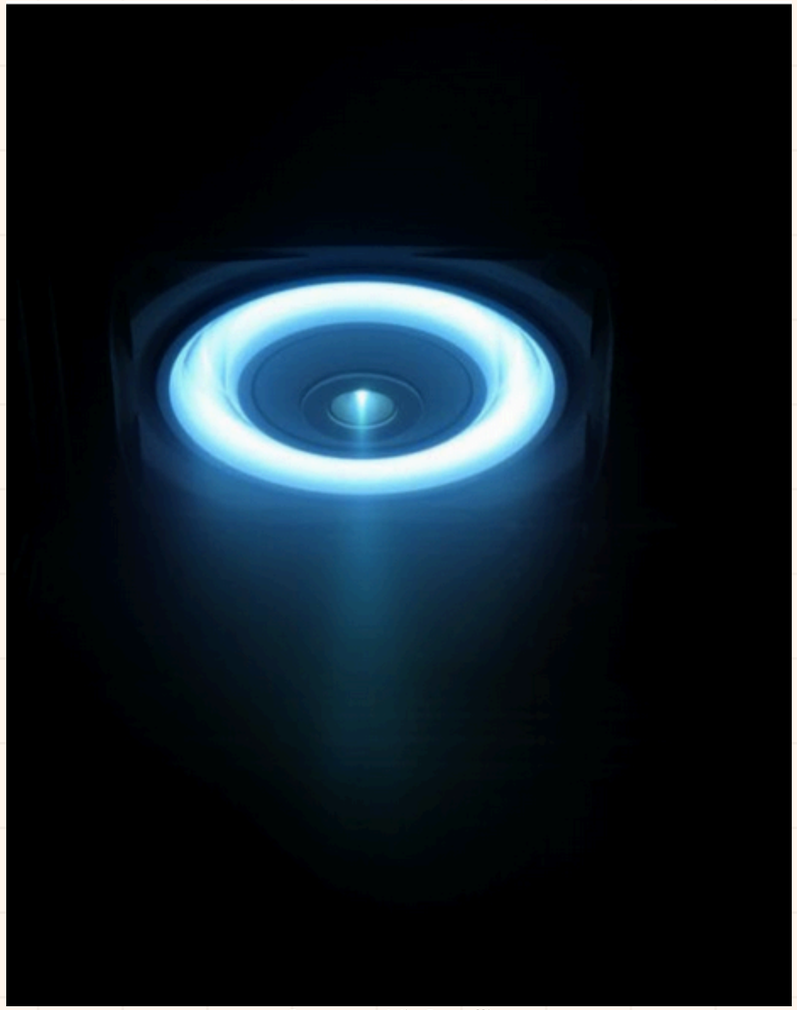
- Chaser satellite utilizing net capturing
- Net is equipped with a Thruster De-orbiting Kit (TDK) to tow targeted debris down to a decay orbit
- Using BHT-8000 thruster
- Low thrust propulsion, T=0.45N, Isp=2210s



Source : ESA

Thruster De-orbiting Kits (TDK's)

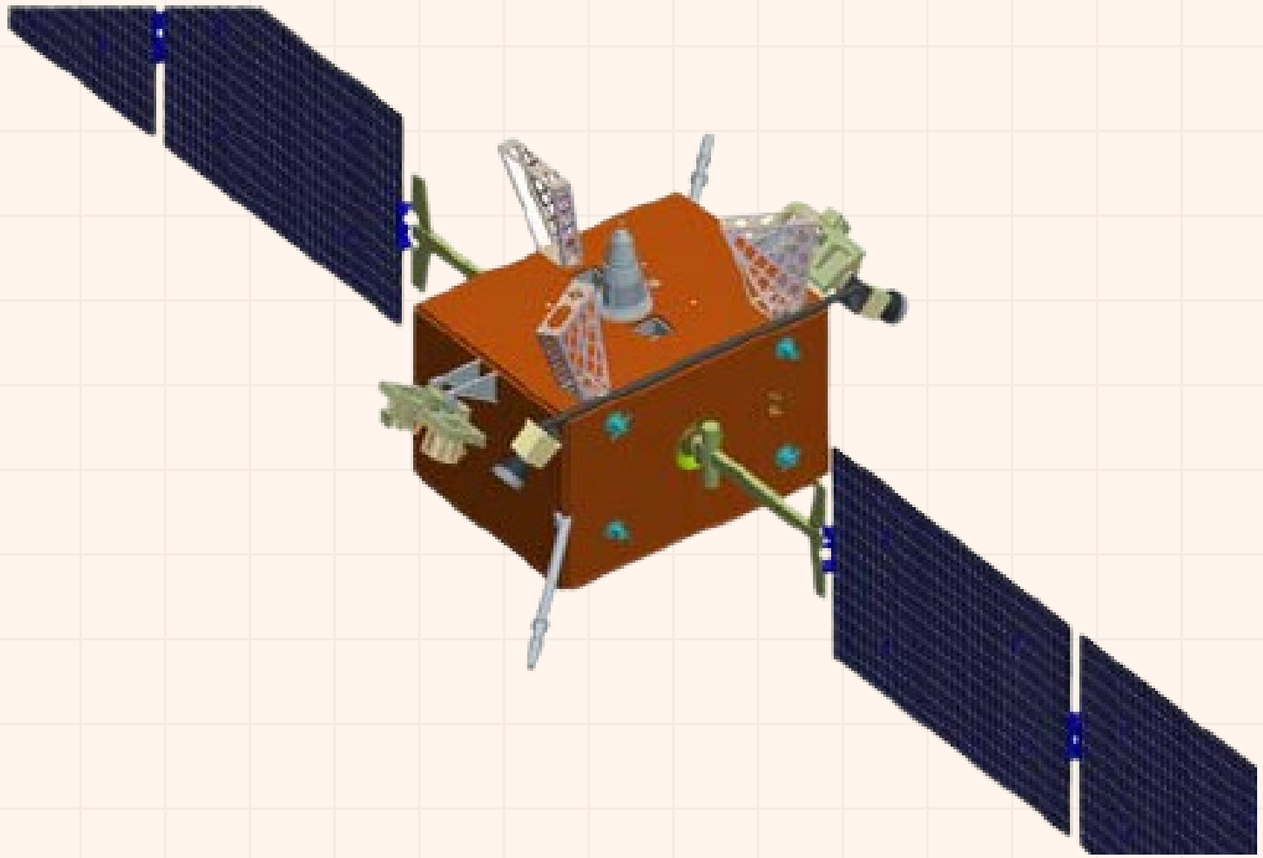
- BHT-8000, Hall Effect thruster (in development)
- Requires 8kW to produce 0.45N of thrust.
- 25kg of Lithium Ion batteries to provide power when not in direct sunlight.
- Solar panels to provide power and recharge the battery when in direct sunlight.
- Total mass : 525kg, including electronics, structural components, solar panels and net.



Source : ViaSatellite

Spacecraft sizing

- Smart-Olev
- Powered by 3 BHT-8000 thrusters, requiring 24kW of power
- Would require:
  - 80 kg of Lithium Ion batteries
  - 108 m² of solar panels → 1210 kg
  - Structural mass of 2000kg
  - 10 TDK’s
- Total mass: 8540 kg



SMART-OLEV, source : Researchgate

Rocket selection

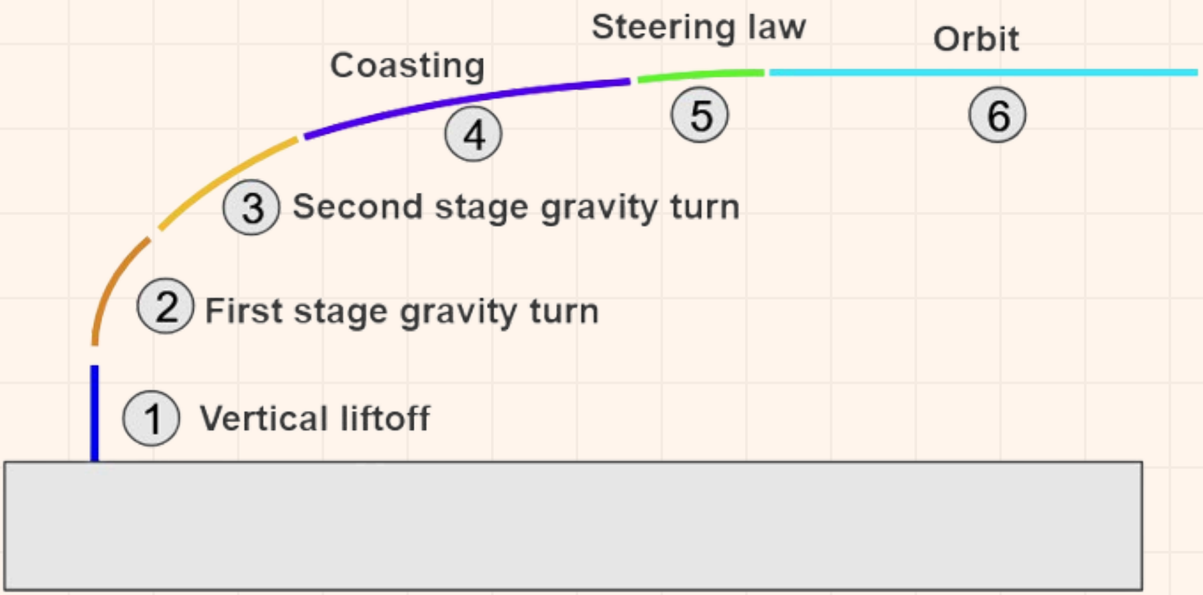
- The **Falcon 9** was chosen as :
  - Cheapest price per kg on the market (2720 $/kg)
  - The most reliable launcher (99%, 356/359 successes)
  - Reusable first stage
  - Up to 22,800 kg to LEO
  - Compatible with Smart-Olev’s size
- Launched from Esrange Space Center (Kiruna)

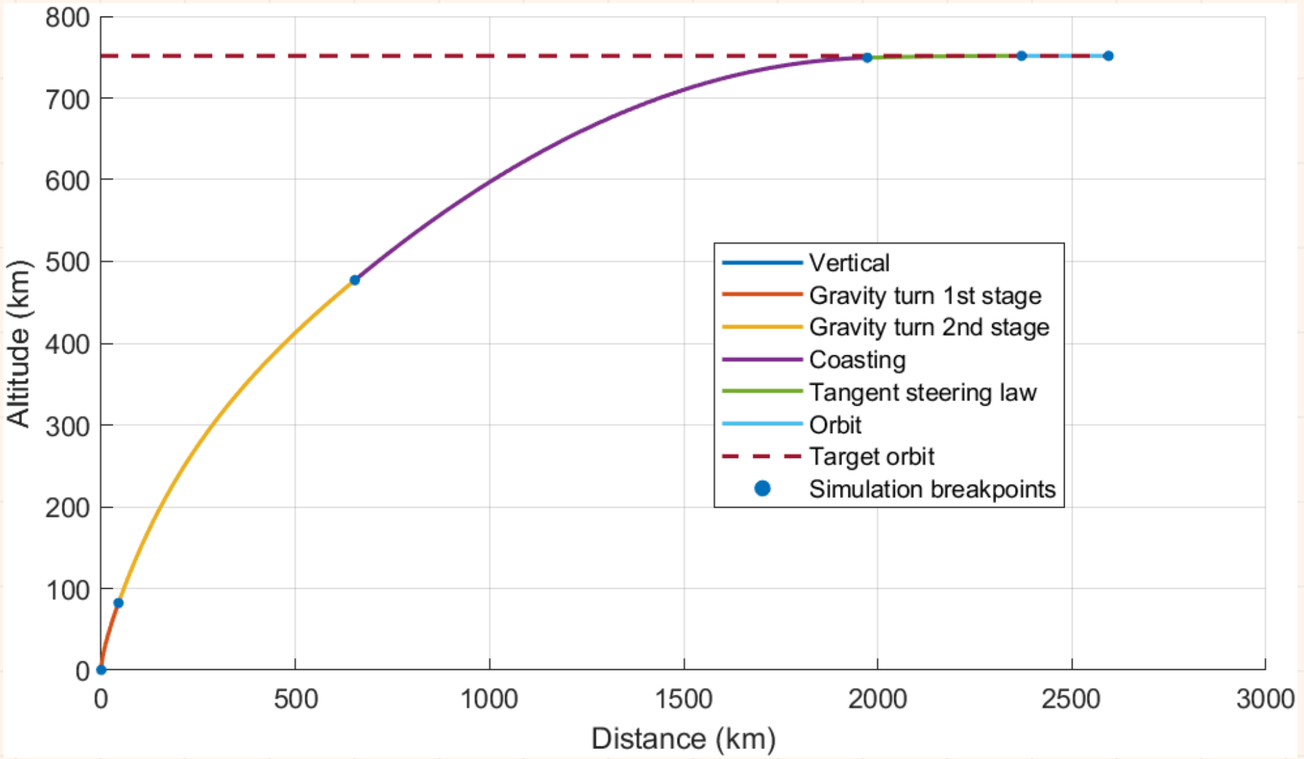


Falcon 9, source : Official SpaceX Photos

Rocket performance/dynamics

- Ascent divided in 6 phases :



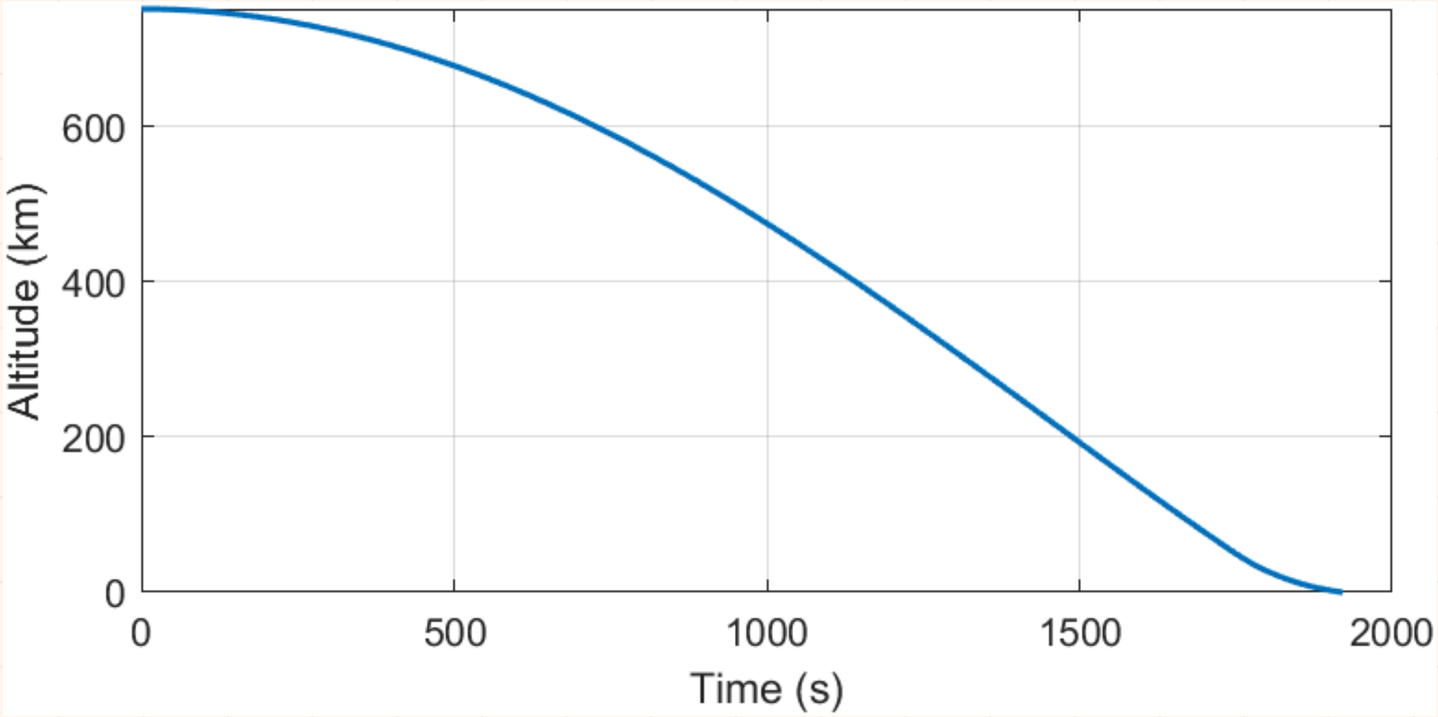


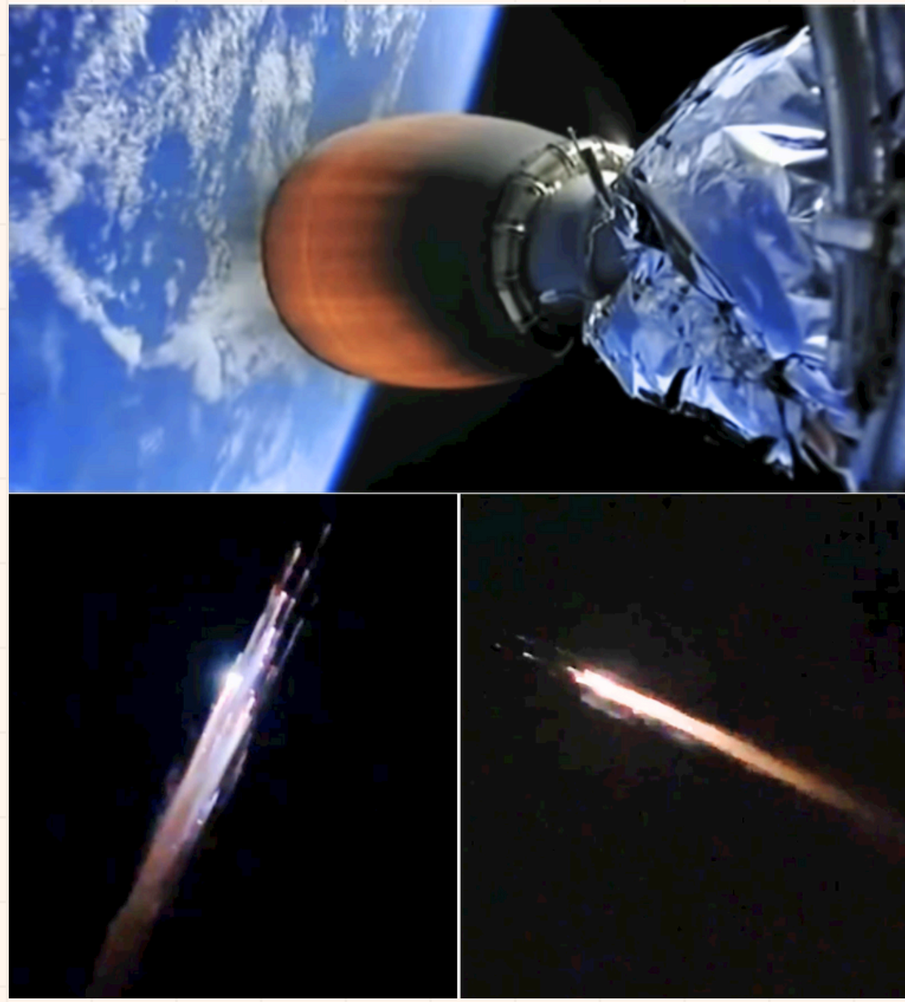
Reaching orbit requires ΔV = 10.56 km/s

Simulation and trajectory optimisation done with MATLAB

Rocket performance/dynamics

- Descent of second stage :
  - **350 kg** of propellant left over in second stage
  - High thrust maneuver to deorbit





Sources : SpaceX, CityNewsEverywhere

Impact 1900s after maneuver

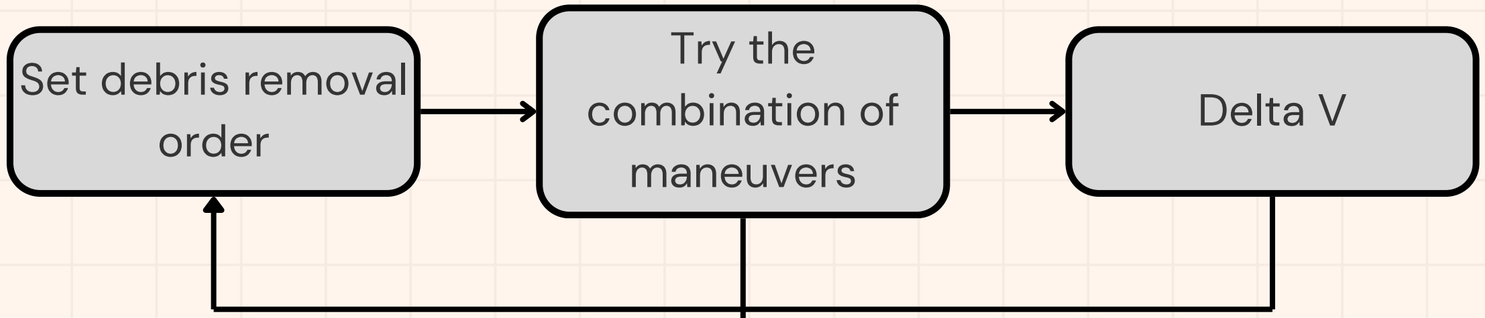
Simulation done with MATLAB

Earth satellite operations

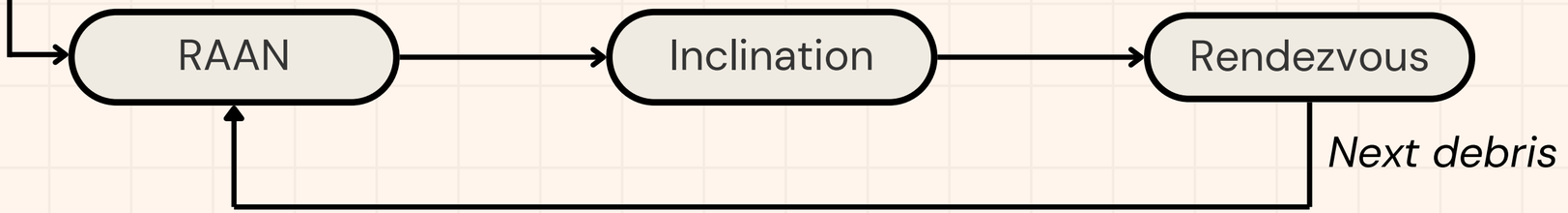
Python program

10 debris to remove : almost **4 million** possibilities of removal order !

How the code works :

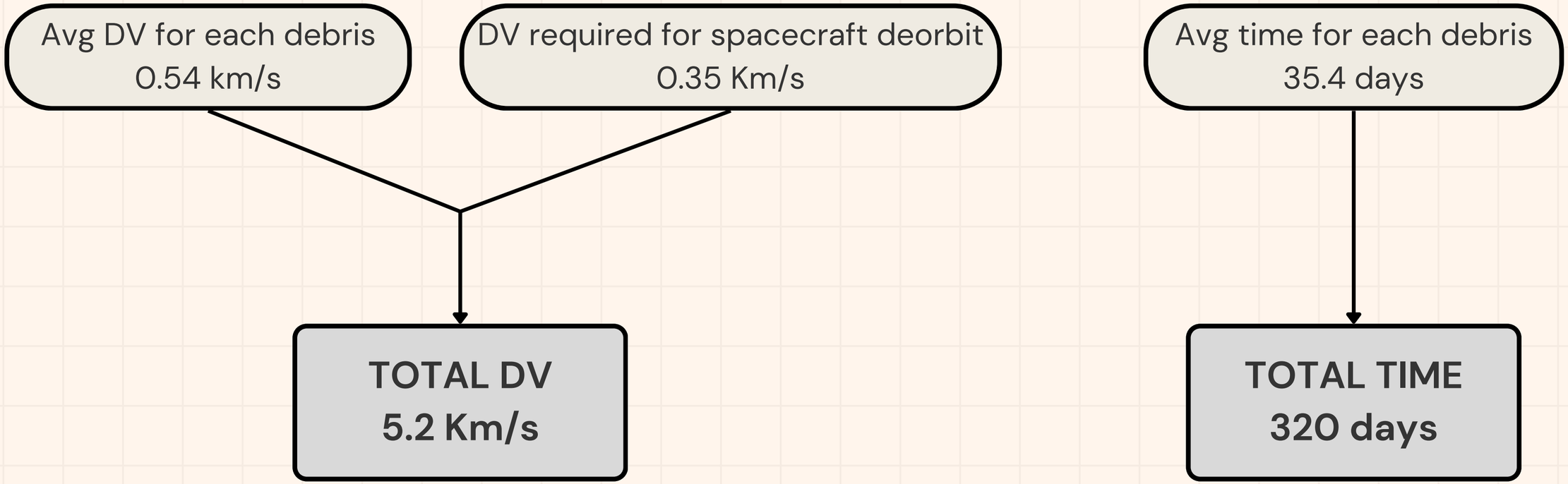


Sequence of maneuvers :



Earth satellite operations

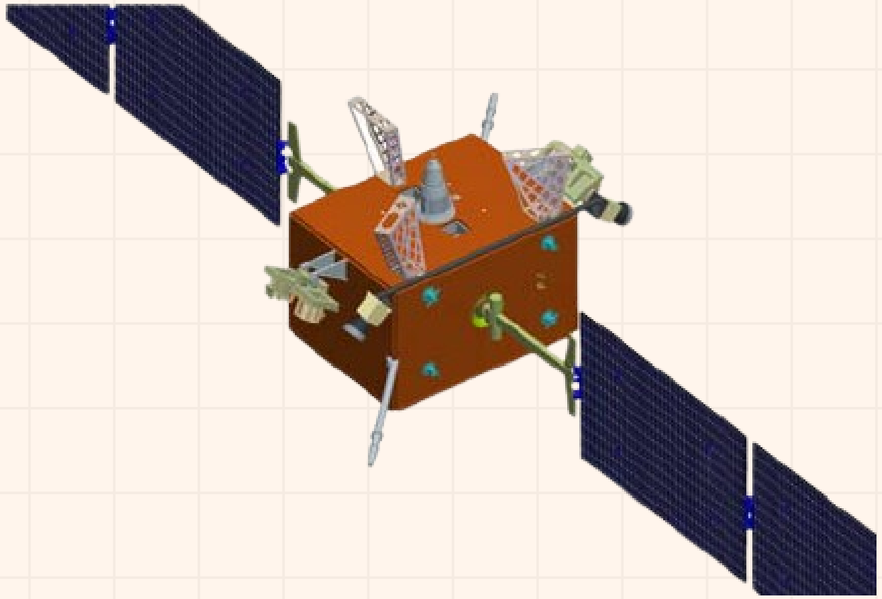
Maneuvers : DV and time for the most effective debris order



Within the constraint of a year-long mission

Cost analysis

- SpaceX service of launching Falcon 9:
  - 62 million USD
- Cost of the spacecraft:
  - 75 million USD
- Total cost:
  - 138 million USD



SMART-OLEV, source : Researchgate



Falcon 9, source : Gunter's space page

Sustainability

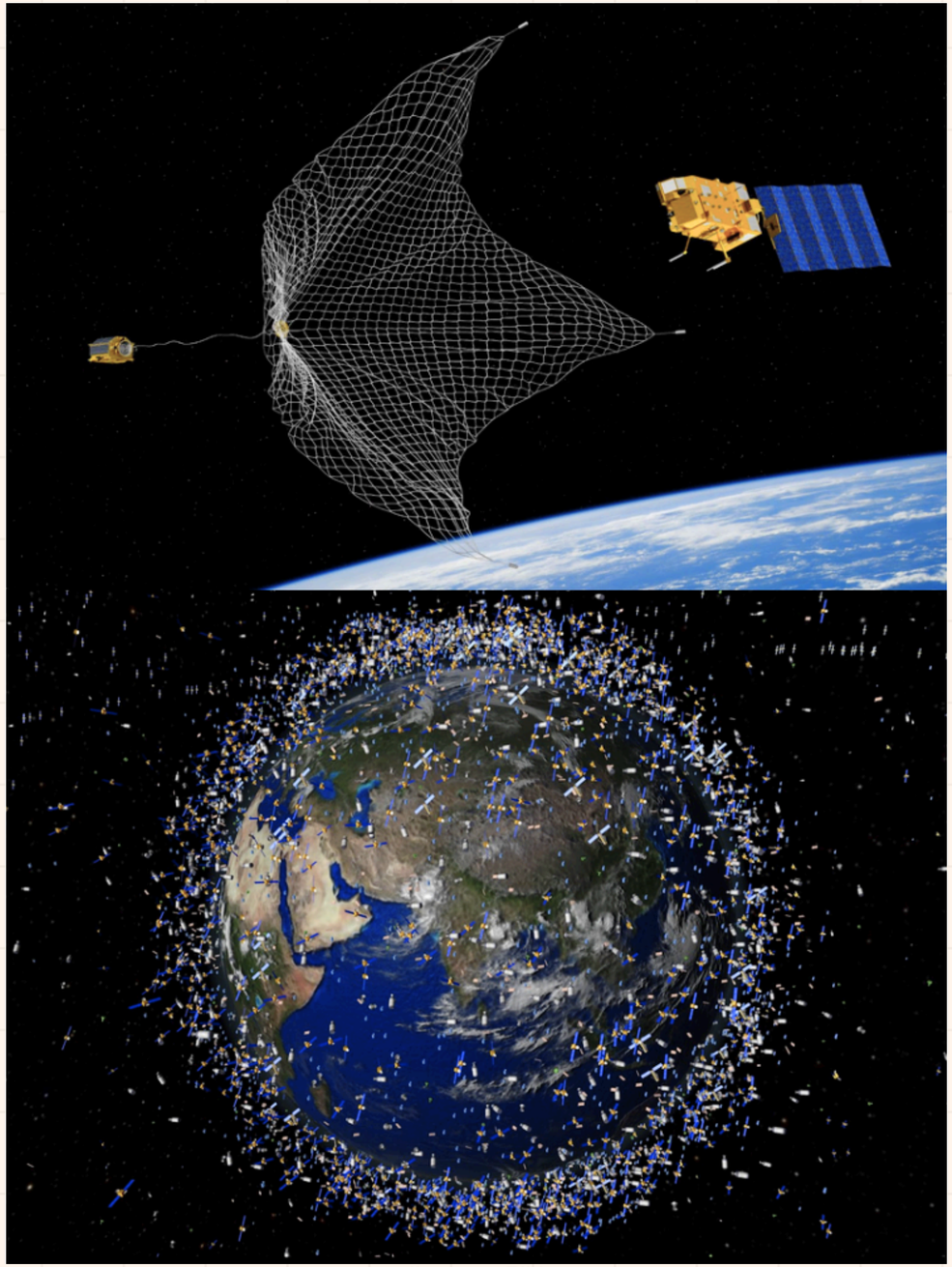
- Emits twice the CO2 of a transatlantic flight
- Only human creation to pollute over 20 km in altitude
- Rocket launches causes emissions high in altitude (damages the ozone layer)
  - Black soot
  - Water vapor (which acts like a green gas in the stratosphere)
  - Nitrogen oxides, etc
- Affects the local population near the launchsite



“What Is Sustainability? And Why Does It Matter?”, source : Green Queen

Conclusion

- Feasibility of a satellite-based active debris removal system is demonstrated
- Able to move 10 large debris to a graveyard orbit within a year
- Mission contributes to sustainability of space operations
- Uses proven, field-tested technologies
- Future work : optimising TDK design, possibility to scale up and target more debris per year



Source : ESA

Thank you for listening !

Any questions ?



Appendix

Earth satellite operations

Assumptions

- Orbits’ eccentricity null
- Low thrust maneuvers (efficiency and control)
- ISO atmosphere model
- Negligible effects of J2

- Natural decayment in 24 years
- Rendezvous :
  - Imposition of an arbitrary derivative of the difference between the true longitudes of debris and spacecraft with respect to time
  - Using one parking orbit